PulsePoint AI RAG Methodology

Transparent project-health scoring for executive delivery oversight

Purpose

PulsePoint AI converts weekly project data into a Red/Amber/Green health status that delivery leaders can trust. The status is calculated from explicit signals, weights, thresholds, and override rules; the reasoning layer explains the fixed result and recommends action.

Core Signals

- Schedule variance: Actual progress compared with expected progress from elapsed time.
- Budget burn ratio: Budget spent relative to delivered work.
- Milestone health: Overdue milestones and near-term commitments at risk.
- Blocker severity: Open risks weighted by severity and age.
- Stakeholder sentiment: PM or client tone extracted from commentary.
- Scope stability: Scope churn applied as a penalty rather than a base score.

Scoring Model

Each signal is normalized to a 0-100 sub-score, where 100 is healthiest. If optional data is missing, its weight is redistributed across available signals. Missing information lowers confidence rather than automatically punishing the project.

Composite = 0.25 Schedule + 0.25 Budget + 0.20 Milestones + 0.20 Blockers + 0.10 Sentiment + Scope Penalty

RAG Interpretation

Green	**75-100**	On track; no material intervention required.
Amber	**50-74**	At risk; needs management attention this week.
Red	**0-49**	Off track; active intervention required.

Override rules prevent severe risks from being hidden by averages. A critical blocker open more than 14 days or budget burn above 1.5x can force escalation.

Data Quality Rules

- Fuzzy field matching handles variants such as % Complete, PercentComplete, and pct_done.
- Bad rows are returned as parse warnings instead of being silently dropped.
- Missing dates or budgets are treated as unknown where appropriate.
- A data confidence percentage is shown beside every RAG status.

Why This Is Trustworthy

The score is reproducible from structured inputs and scoring_config.yaml. The reasoning layer cannot change the score; it can only explain it using read-only evidence. Weekly reports include sub-scores, confidence, warnings, risks, actions, and a reasoning trace.